

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456940.901 +/- 0.075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.05
Transit depth (Rp/Rs)^2	3.0 +/- 1.74 [%]
Orbital Inclination (inc)	84.25 +/- 2.88
Airmass coefficient 1 (a1)	1.502 +/- 0.021
Airmass coefficient 2 (a2)	-0.278 +/- 0.071
Scatter in the residuals of the lightcurve fit is	1.37 %
Best Comparison Star	None
Optimal Aperture	3.9
Optimal Annulus	12.78
Transit Duration (day)	0.021 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.05 vs 0.1404 (deviation 0.65 σ)
Duration fit vs published	0.021 d vs 0.0754 d (deviation 2.27 σ)
Mid-time O-C	54.2 min
Depth SNR	3.5
Residual scatter	1.37 %

Light curve

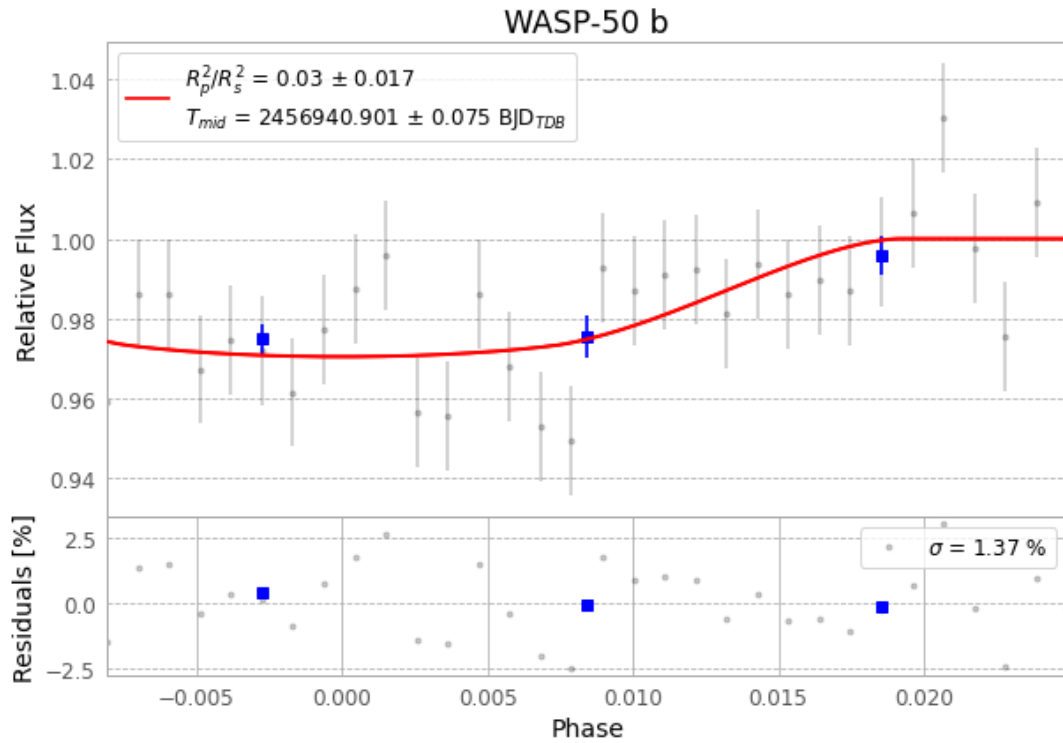


Figure 1 — FinalLightCurve_WASP-50 b_2014-10-10.png (SHA-256 487d37e0f4da20c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [532, 411], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 77cbb18debc6b2be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

32 FITS frames (21 MB), WASP-50141010090612.FITS → WASP-50141010103913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-141010.log (a877ddcbb4e6...), FinalLightCurve_WASP-50 b_2014-10-10.png (487d37e0f4da...), FinalLightCurve_WASP-50 b_2014-10-10.csv (42ecea152a4b...)

Compute resources

Wall time 85.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 04:19Z.